6CLOUDCOM Final Project

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Parejas, Arron Kian M.
Tongol, Jenica Sarah B.

CS-301

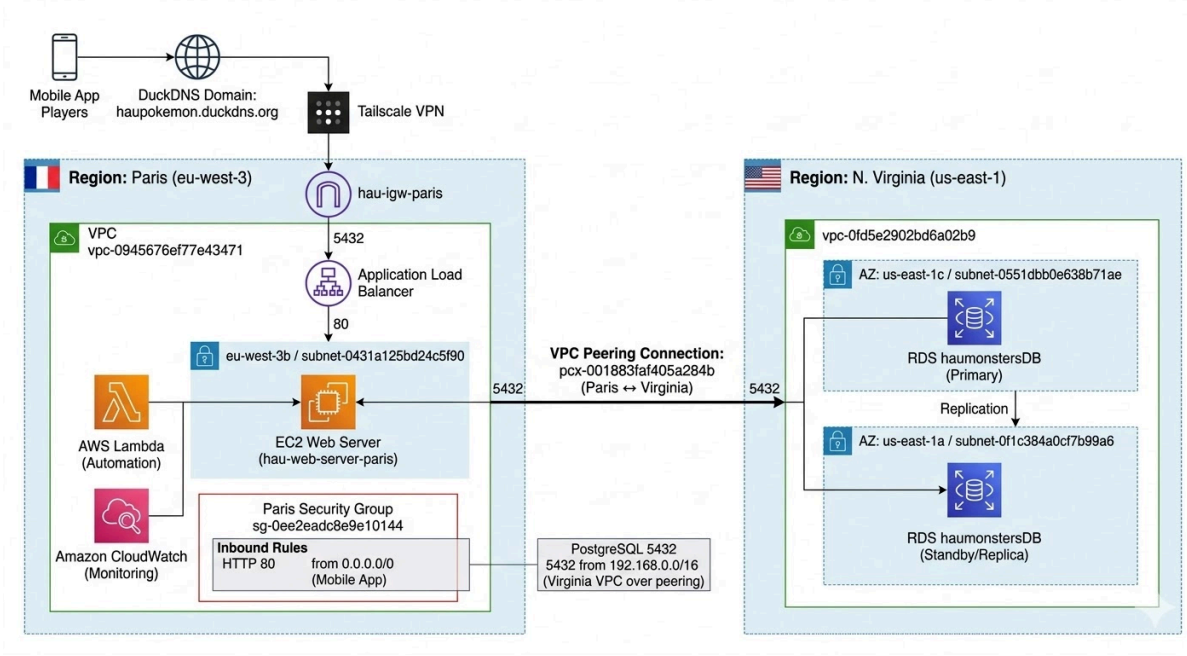
March 26, 2026

HAUPokemon Monster's App

Team Task List & Contributions

Team Member	Project Role	Core Contributions
Jimenez, Graciella Mhervie D.	Project Manager	Handled project management, comprehensive documentation, rubric compliance validation, and team coordination.
Marquez, Jian Kalel D.	Mobile Developer	Developed the Flutter mobile app, integrated custom UI/UX, built hardware triggers (flashlight/alarm), and connected AWS Lambda EC2 automation.
Parejas, Arron Kian M.	Cloud Engineer	Architected the AWS backend, including Multi-AZ PostgreSQL RDS, Application Load Balancer, EC2 Web Server, VPC Peering, and Tailscale VPN.
Tongol, Jenica Sarah B.	UI/UX & Logic	Handled UI/UX design components, coordinate/distance mapping logic, and assisted with the responsive application layout.

Team Task List & Contributions



The HAUPokemon cloud infrastructure is designed for high availability, cross-region security, and cost optimization. User traffic originates from the Mobile App and connects securely via the Tailscale VPN through our custom DuckDNS domain. Traffic enters the Paris Region (eu-west-3) via an Internet Gateway and routes through an Application Load Balancer on Port 80 to ensure high availability before reaching the EC2 Web Server. To optimize costs, serverless components (AWS Lambda and Amazon CloudWatch) are deployed in the Paris VPC to automate and monitor the EC2 instance's start/stop states based on app usage.

For secure data storage, the Paris EC2 server connects via a cross-region VPC Peering Connection to the North Virginia Region (us-east-1). The backend safely executes SQL queries over TCP Port 5432 to a Multi-AZ PostgreSQL RDS deployment. This multi-AZ setup ensures continuous database availability with an active Primary database in one availability zone and an automated Standby/Replica in another for immediate failover.

AWS Pricing & Cost Estimate

Project Name: HAUPokemon Monster's App
Pricing Model: AWS Free Tier Credits
Current Billed Period: March 1 - March 26, 2026
Currency: USD

Service Category	Infrastructure Details	Region	Estimated Cost	Actual Cost
Domain & VPN	DuckDNS (haupokemon.duckdns.org) + Tailscale	Global	**$0.00**	**$0.00**
Virtual Private Cloud	Cross-region VPC Peering & Data Transfer	Cross-Region	**~$0.20**	**$5.85**
Compute (Web Server)	Amazon EC2 t3.micro + EBS Storage	Paris (eu-west-3)	**~$9.50**	**$5.03**
Database Compute	Amazon RDS PostgreSQL Multi-AZ (db.t3.micro)	N. Virginia (us-east-1)	**~$31.00**	**$3.94**
High Availability	Elastic Load Balancing (ALB)	Paris (eu-west-3)	**~$22.50**	**$2.06**
DNS	Amazon Route	Global	**$0.00**	**$0.50**

Management	53			
Automation	AWS Lambda (EC2 Start/Stop script)	Paris (eu-west-3)	**$0.00**	**$0.00**
		TOTALS:	**~$63.20**	**$17.38**

To demonstrate real-world cloud cost management, our Cloud Engineer imposed a strict $50.00 USD credit budget for this entire project. Through aggressive resource optimization, we successfully deployed a highly available, multi-region architecture (Paris to North Virginia) while only consuming $17.38 of our allocated budget.

By outsourcing our custom domain to DuckDNS and deploying Tailscale's free tier for our VPN network, we effectively reduced our security and routing overhead to zero. Furthermore, to prevent runaway compute costs on our web server, we implemented serverless computing via AWS Lambda to manage the ON/OFF states of the Paris EC2 instance, ensuring we only incurred charges for active compute hours. This strategy allowed us to comfortably afford a Multi-AZ PostgreSQL RDS deployment and an Application Load Balancer without breaking our self-imposed limit.

IAM Credentials for Grading

To verify our cloud infrastructure, VPC Peering, and Database schemas, please use the following IAM credentials to access our AWS Management Console.

- **AWS Login URL:** https://285666934754.signin.aws.amazon.com/console
- **IAM Username: adminproject**
- **IAM Password: Cloudcomp2026**
- **Assigned Policies:** AdministratorAccess, AmazonEC2FullAccess, AmazonVPCFullAccess, AmazonRDSFullAccess

Important Links & Assets

Resource	Direct Link
GitHub Repository	https://github.com/darknecrocities/Pokemon-Hau
UI/UX Figma Design	https://www.figma.com/design/JWg4tYXqCthe9QvidZYsty/6CLOUDCOM_HAUPokemon_UIUX?t=y1oauhwIqSetJkuW-0